

AI Infrastructure Leverage Monitor

Issue 1: Baseline — How much leverage sits behind the AI buildout, and where

Kilambi Research

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Scope. This weekly program measures the debt, credit, leverage, contractual obligations and off-balance-sheet financing supporting the global AI infrastructure boom: hyperscalers and semiconductor firms, data-center and neocloud operators, private frontier-model companies, the banks that lend to them, and the private-credit, insurance, BDC and structured-credit vehicles around them. Every major number cites its primary document. Company disclosures are separated from researcher calculations. Unknown exposure is reported as unknown. Data as of 26 September 2026 unless stated.

1 Executive summary

This first issue establishes the baseline. Fourteen public borrowers carry \$671 billion of balance-sheet debt in the latest filings, led by Amazon (\$132.2B), Oracle (\$125.3B) and Alphabet (\$100.2B). The disclosed fixed commitments behind that debt are far larger: Alphabet reports \$707B of fixed or guaranteed commitments, Meta \$349B of non-cancelable contractual commitments, Microsoft \$329B of uncommenced data-center leases, and Nvidia \$279B of supply and capacity commitments. Where the full five-tier ladder is measurable, economic fixed obligations run 4 to 9 times reported debt. These commitments are disclosed in filings, not hidden; the genuinely opaque exposures are the private labs (OpenAI discloses one \$4.7B undrawn revolver and nothing else on its balance sheet; Anthropic discloses no debt facility at all) and the manager-level private-credit books, of which public BDC schedules capture only \$520M. Bank capital is ample (CET1 10.3% to 17.2%) and only one bank in thirteen, SMFG, quantifies AI-related exposure (\$24.9B, 2.3% of its total). Refinancing risk concentrates in the neoclouds: CoreWeave amortizes \$20.6B of principal across 2026–2030 against \$35.1B of net debt, while hyperscaler maturities sit mostly beyond 2030. The evidence supports the claim that economic leverage materially exceeds headline debt; it does not support the claim that the excess is concealed.

2 What changed this week

This is the baseline issue, so there is no prior vintage to difference against. What this section establishes:

1. The master table is populated: 70 entities, 90 obligations (57 primary, 33 memo rows excluded from aggregates), 55 sources, all append-only under vintage 2026-09-27.
2. Five-tier leverage baselines are computed for 14 public borrowers at the latest balance-sheet date and at FY2022 where clean history exists.

3. Lender baselines are set: CET1, loan books, unfunded commitments and disclosed AI exposure for 13 banks; verified BDC portfolio positions for five BDCs.
4. The maturity and commitment wall is built through 2035 from disclosed schedules, with gaps marked where disclosure does not permit allocation.
5. Revisions box: none. This is the first vintage.

3 Corporate leverage dashboard

3.1 Reported debt

Balance-sheet debt (Tier A, strict GAAP) totals approximately \$649B across the 14 public borrowers at the latest filing dates (researcher sum of carrying amounts; fiscal periods differ across companies).

Table 1: Balance-sheet debt, latest filing (USD billions)

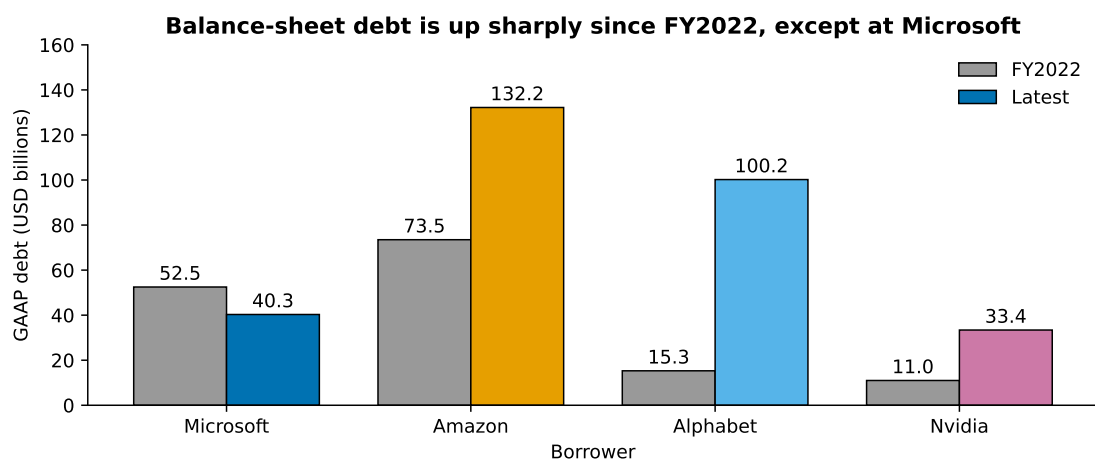
Borrower	Debt	Period end	Filing
Amazon	132.2	30 Jun 2026	10-Q filed 31 Jul 2026
Oracle	125.3	31 Aug 2026	10-Q filed 11 Sep 2026
Alphabet	100.2	30 Jun 2026	10-Q filed 23 Jul 2026
Meta	83.7	30 Jun 2026	10-Q filed 30 Jul 2026
Broadcom	59.4	2 Aug 2026	10-Q filed 10 Sep 2026
Microsoft	40.3	30 Jun 2026	10-K filed 29 Jul 2026
CoreWeave	35.1	30 Jun 2026	10-Q filed 12 Aug 2026
Nvidia	33.4	26 Jul 2026	10-Q filed 26 Aug 2026
Equinix	19.7	30 Jun 2026	10-Q filed 29 Jul 2026
Digital Realty	18.6	30 Jun 2026	10-Q filed 31 Jul 2026
Nebius	8.5	30 Jun 2026	6-K filed 12 Aug 2026
TeraWulf	5.2	30 Jun 2026	10-Q filed 5 Aug 2026
Applied Digital	5.0	31 May 2026	10-K filed 29 Jul 2026
Core Scientific	4.3	30 Jun 2026	10-Q filed 28 Jul 2026

Debt growth since FY2022 is available on a clean basis for four borrowers. Amazon rose 79.8% (\$73.5B to \$132.2B); Nvidia rose 204.7% (\$11.0B to \$33.4B); Alphabet rose 554.2% (\$15.3B to \$100.2B), though both FY2022 bases are long-term debt only, so the rate is overstated while the direction is not in doubt. Microsoft is the only one of the four to reduce balance-sheet debt, down 23.3% (\$52.5B to \$40.3B), even as its off-balance-sheet commitments grew far faster.

3.2 The five tiers

The program measures five cumulative tiers: (a) strict GAAP debt; (b) plus finance leases; (c) plus all recognized lease liabilities; (d) plus uncommenced non-cancelable leases; (e) plus unavoidable purchase, take-or-pay and compute commitments and probability-weighted guarantees. Where the ladder is complete, the step from (a) to (e) is dominated by items (d) and (e).

Alphabet reaches \$913.0B at Tier E against \$100.2B of GAAP debt; its \$707B of fixed or guaranteed commitments (Note 10, 10-Q for the quarter ended 30 June 2026, filed 23 July 2026) are 7.1 times its reported debt. Microsoft reaches \$686.5B at Tier E against \$40.3B of debt; its



Data as of latest filings (June to July 2026).

Source: SEC EDGAR, company 10-Q and 10-K filings. Accessed 27 Sep 2026.

Note 1: Alphabet and Nvidia FY2022 bases are long-term debt only (current portion not extracted), so their growth rates are overstated; the direction is not in doubt.

Figure 1: Balance-sheet debt trajectories, FY2022 to latest filing.

\$329.1B of uncommenced data-center leases (Note 11, FY2026 10-K filed 29 July 2026) are 8.2 times its reported debt. Amazon reaches \$509.3B at Tier E. CoreWeave, the most levered neocloud on balance sheet, reaches \$87.1B at Tier E against \$35.1B of debt, with the increment coming almost entirely from leases.

Blanks in the ladder are deliberate and documented. Meta stops at Tier A because its Q2 10-Q does not disclose finance-lease liabilities (the FY2025 10-K figure of \$1,184M was not carried forward). Nvidia stops at Tier A (finance leases not separately disclosed). Oracle stops at Tier C because its \$260B of uncommenced data-center leases, disclosed in the May 2026 10-K, were not repeated in the August 10-Q and were not carried forward. These commitments remain recorded as obligations; they are kept out of the cumulative ladder only.

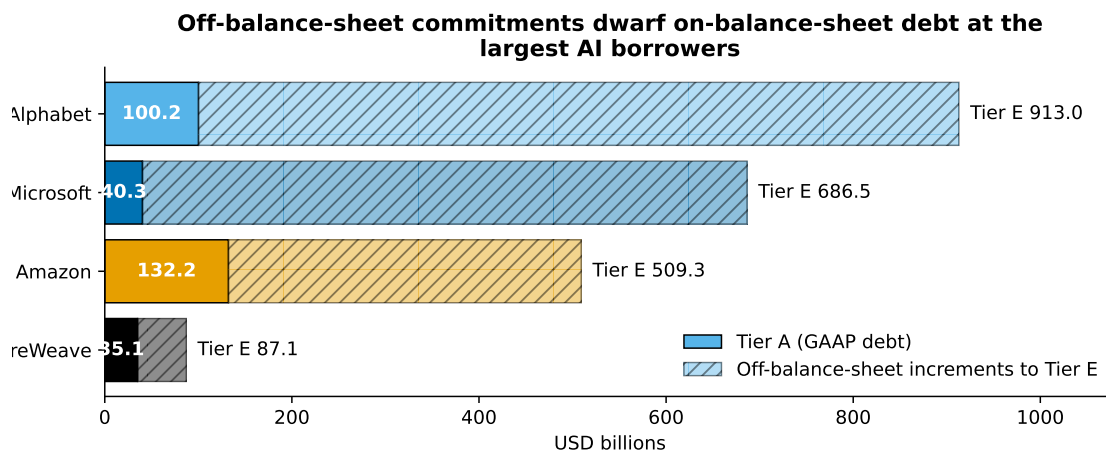
3.3 Contingent and debt-like commitments

Three guarantee-class items deserve separate attention because they are large, disclosed, and sit entirely outside reported debt. Nvidia signed a capped guarantee of \$105B in August 2026 backing approximately 4.25 GW of SB Energy buildout for an OpenAI affiliate in Ohio (subsequent event to the 26 July 2026 balance sheet; maximum gross exposure, not expected loss). That is 3.1 times Nvidia's own GAAP debt. Broadcom carries a \$29B AI-rack lease backstop plus a customer option to require issuance of \$42B in convertible notes. Oracle guaranteed up to \$3.3B of lessor borrowings tied to one of its uncommenced leases.

The largest disclosed fixed-commitment aggregates, each an aggregate portfolio rather than a single contract and never summed with each other or with debt, are shown in Figure 3.

3.4 The private labs

OpenAI's only company-disclosed debt instrument is its revolving credit facility, expanded to approximately \$4.7B at the March 2026 close and undrawn at close (company announcement; the October 2024 \$4.0B facility is the same economic facility). OpenAI discloses no total debt, no



Data as of 30 Jun 2026.

Source: SEC EDGAR, company 10-Q filings. Accessed 27 Sep 2026.

Note 1: Tiers are cumulative per program definitions; each bar shows Tier A plus the incremental off-balance-sheet amount to reach Tier E. CoreWeave Tier E equals Tier D (no purchase commitments disclosed beyond leases).

Figure 2: Tier A (GAAP debt) against Tier E (economic fixed obligations) for the four borrowers with complete ladders.

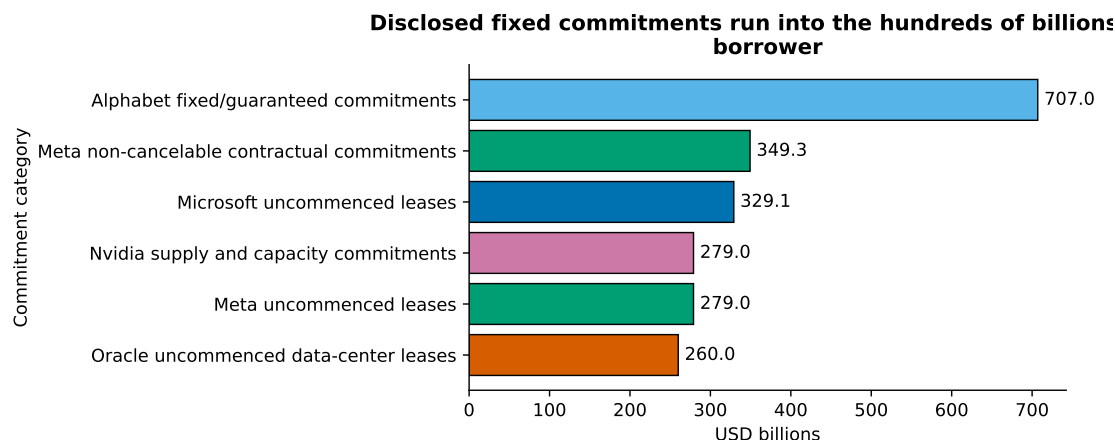
leases, no purchase-obligation schedule, no cash position and no covenants. Anthropic has no primary-source confirmed debt facility; the reported \$15B revolver is press-only and excluded from primary aggregates. Anthropic's two company-stated commitments are more than \$100B over ten years to AWS (securing up to 5 GW) and \$30B of Azure compute purchases. Counterparty filings add: Microsoft has committed \$13B to OpenAI with \$11.7B funded at December 2025 and approximately 27% as-converted ownership after the October 2025 recapitalization (Microsoft 10-Qs); Amazon invested \$8B in Anthropic convertible notes across Q3 2023 to Q4 2025 (Amazon 10-Q). The \$500B Stargate figure is stated as an intention, not a liability; the \$300B-plus Oracle partnership figure is described as partnership value over five years, not a proved payment obligation. Both are memo rows.

4 Lender exposure dashboard

4.1 Banks

Capital is ample across the cohort. CET1 ratios for the June 2026 reporting periods run from 10.26% (Wells Fargo) to 17.2% (Crédit Agricole Group), all well above regulatory minimums.

On AI-specific exposure, the finding is mostly negative, and the negative is the finding. Eight of thirteen banks made substantive AI-related disclosures (commentary, platforms, pipeline remarks), but only one, SMFG, published a quantified exposure figure: JPY 4.0 trillion at 31 March 2026 (SMBC consolidated, managerial-accounting basis; FY2025 investor presentation, May 2026), of which JPY 2.6tn is data centers and JPY 1.3tn software. Converted at the official SMBC fixing of JPY 160.90 per USD, that is \$24.9B (researcher calculation), equal to 2.3% of SMFG's JPY 177tn total exposure. The funded versus unfunded split is not disclosed, and the managerial-accounting basis is not comparable to loan-book disclosures. SMFG's Q&A states roughly 80% of data-center long-term offtakers are hyperscalers and that underwriting standards are becoming more selective. Every other bank in scope discloses no AI-specific exposure figure. That does not mean the exposure is zero; it means it cannot be measured from primary sources.



Data as of latest filings (May to July 2026).

Source: SEC EDGAR, company 10-Q and 10-K filings. Accessed 27 Sep 2026.

Note 1: Each bar is an aggregate portfolio disclosed in filings, not a single contract; these commitments are not added to debt. Oracle figure is from the May 2026 10-K and was not repeated in the August 10-Q.

Figure 3: Largest disclosed fixed-commitment aggregates per borrower (USD billions).

4.2 Private credit, BDCs, insurers, structured credit

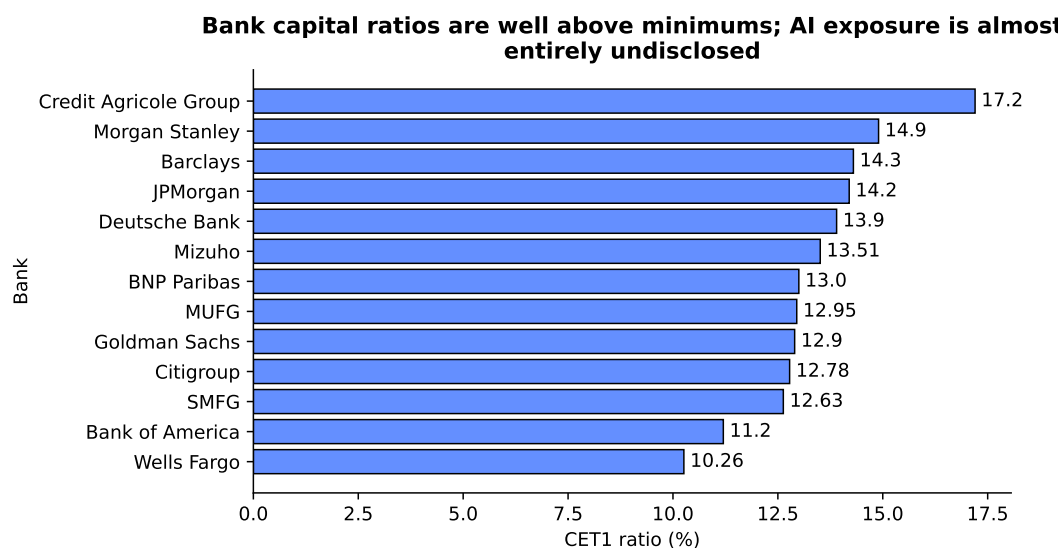
Filed fair-value positions at 30 June 2026 across five BDCs total \$520.4M funded: Ares Capital's \$195.9M and \$40.1M Vantage Data Centers positions and \$83.7M EdgeConneX; Blackstone Private Credit Fund's \$159.3M CyrusOne revolving warehouse and \$21.2M Emerald JV Vantage position; Carlyle Secured Lending's \$16.6M and \$1.9M CoreWeave SPV positions; HPS Corporate Lending Fund's \$0.9M CoreWeave DDTL V; Blue Owl Capital Corp's \$0.9M Nscale. Known unfunded commitments total \$290.1M. All BDC rows are memo in the master table: they are lender-side views at filed fair value of SPV debt that consolidates into borrower balance sheets already recorded, and entering them into borrower aggregates would double count.

The \$520.4M is three orders of magnitude smaller than the borrower fixed-commitment aggregates in Section 3. That states the scale of what public BDC schedules can and cannot show. No primary source quantifies any manager's total AI-infrastructure credit book; CLOs, private funds, separately managed accounts and insurance general accounts are not quantified in the primary material. Apollo's Intel Fab 34 position is excluded: it was an equity joint venture repurchased by Intel in April 2026, not credit. A strict managed-versus-retained discipline is applied throughout: AUM is third-party capital, not the manager's risk, and is never presented as exposure.

No data-center or equipment ABS issuance was identified for any borrower in the cohort from 10-K and 10-Q note text, though absence cannot be fully established without exhibit-level indenture review. Rating-agency presale material was located for third-party data-center vehicles (Vantage, CyrusOne); spreads and subordination from those presales will enter the dashboard when the first refinancing calendar is published.

5 Maturity and commitment wall

Borrower-side principal maturities are concentrated well beyond 2030 for the hyperscalers: Microsoft \$34.4B due thereafter; Meta \$73.8B thereafter; Oracle \$90.3B thereafter; Broadcom \$44.9B thereafter. The wall inverts for the neoclouds. CoreWeave's \$35.6B of gross principal amortizes \$4.4B in the remainder of 2026, \$6.2B in 2027, \$4.4B in 2028, \$2.4B in 2029 and \$3.2B in 2030.



Data as of June 2026 reporting periods.

Source: company regulatory filings and earnings releases. Accessed 27 Sep 2026.

Note 1: Only SMFG discloses a quantified AI-related exposure (JPY 4.0tn); all other banks disclose none.

Figure 4: CET1 ratios, June 2026 reporting periods (filed figures).

Near-term fixed cash requirements visible in the wall include Meta's \$53.5B of commitments due in the remainder of 2026, Broadcom's \$52.7B of purchase commitments due in 2027, and Nvidia's \$120B of aggregate future commitments due in the remainder of FY2027.

Covenant pressure points identified from primary documents: CoreWeave's revolver accordion carries a 6.00x total net leverage test (4.00x secured net) with DDTL DSCR sizing tests described only qualitatively, so distance to breach cannot be measured from filings. Core Scientific's 2031 secured notes amortize at 11.5% per year with an uncapped parent completion guarantee. TeraWulf's 2030 secured notes carry an excess-cash-flow mandatory repurchase with no-new-debt and no-distribution negative covenants. Digital Realty's indenture package caps leverage at 60%, secured leverage at 40%, and requires 1.50x coverage with 150% unencumbered-asset coverage. Applied Digital's 2030/2031 secured notes limit additional indebtedness and restricted payments, with parent completion guarantees. Equinix's notes carry no numeric covenants.

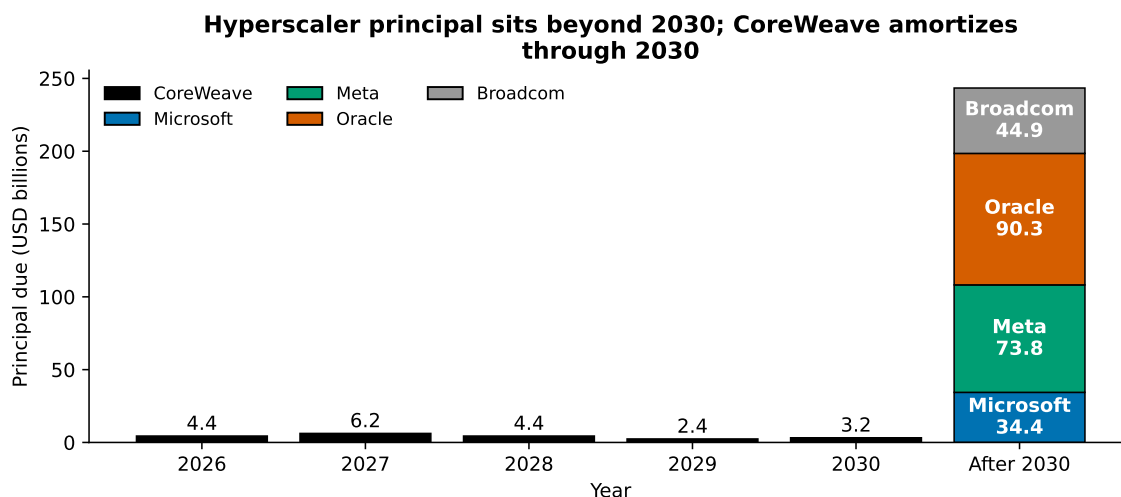
The lender-side refinancing calendar cannot yet be built: no bank or private-credit manager discloses a maturity schedule for its AI or data-center loan book. That is recorded as a gap, not filled by assumption.

6 Rotating focus: the five leverage tiers, explained

Each tier answers a different question, and the tiers are not interchangeable.

Tier A, strict GAAP debt, answers what the auditors recognize as borrowing. It is the only tier comparable across companies without judgment, and the only tier for which FY2022 history is clean. Its weakness is scope: it excludes everything the accounting standards deliberately keep off the balance sheet.

Tier B adds finance leases. Economically these are secured borrowings with the asset as collat-



Data as of latest filings (June to August 2026).

Source: SEC EDGAR, company 10-Q and 10-K filings. Accessed 27 Sep 2026.

Note 1: Summary view: CoreWeave annual principal 2026-2030 plus the four largest named post-2030 principal concentrations. Modest hyperscaler principal due before 2031 and CoreWeave amounts due after 2030 are disclosed in the maturity CSV but omitted here.

Figure 5: Maturity and commitment wall: hyperscaler principal sits beyond 2030 while CoreWeave amortizes through 2030.

eral; GAAP agrees, which is why they sit on the balance sheet. The adjustment is uncontroversial. Its weakness in this vintage is disclosure: several borrowers do not break finance leases out at the latest quarter.

Tier C adds all recognized lease liabilities, including operating leases. Since ASC 842, operating leases are on the balance sheet, so for most borrowers Tier C is close to Tier B plus a disclosed number. It is economically equivalent to debt for fixed, non-cancelable leases: the company must pay or default. The residual difference is renewal options and variable payments, which the standards exclude.

Tier D adds uncommenced non-cancelable leases: signed data-center leases that have not started and therefore sit nowhere on the balance sheet. Microsoft's \$329.1B and Oracle's \$260B are the large examples. These are economically debt-like to the extent commencement is certain, which for build-to-suit data-center shells with contractual conditions is high but not 1.0. The program records them at notional in Tier D and discusses commencement risk in prose rather than haircutting the number silently.

Tier E adds unavoidable purchase, take-or-pay and compute commitments plus probability-weighted guarantees. Alphabet's \$707B fixed-or-guaranteed commitments dominate its Tier E. The probability weights are the only judgmental input in the ladder and are documented per item in `tier_e_weights.md`. Nvidia's \$105B SB Energy guarantee and Broadcom's \$29B backstop are carried at maximum notional with weight 1.0 this vintage, flagged as maximum exposure rather than expected loss. Tier E is the right measure of "what must be paid if the AI thesis is wrong"; it is not the right measure of default risk, because purchase commitments can often be renegotiated while bonds cannot. Both statements are true and the report keeps them separate.

7 Stress and transmission watch

Scenarios are monitored, not forecasted. For each, the watch states where pressure would emerge first given current evidence.

20–40% lower compute pricing. First pressure at the neoclouds: CoreWeave’s DDTL DSCR sizing tests and 6.00x leverage test are the nearest covenants, and its \$20.6B of 2026–2030 amortization cannot be repriced downward. Hyperscaler impact would run through utilization of uncommenced leases, not covenants.

Major customer failure or contract renegotiation. CoreWeave’s revenue concentration (majority with one hyperscaler, per its 10-Q) makes this the single largest idiosyncratic risk in the dataset. Applied Digital’s lease assignment to CoreWeave SPV CCAC VIII (March 2026, with springing parent guaranty and a \$50M letter of credit) shows how customer risk is already being collateralized one layer down.

Refinancing rates 200–400 bp higher. Hyperscaler maturities beyond 2030 make this a 2030s problem for them; for CoreWeave, TeraWulf and Core Scientific, whose facilities amortize or mature before 2032, it is a current-cycle risk. The lender-side calendar is undisclosed, so the corresponding bank and private-credit refinancing pressure cannot be located.

Falling GPU and data-center residual values. This strikes collateral values first: equipment-backed facilities, any borrowing-base structures, and ABS collateral pools. No borrowing-base breach can be confirmed from current disclosure; the watch item is new ABS presale subordination levels.

Second-order transmission. The plausible chain runs: neocloud covenant pressure, lender forbearance or restructuring, tighter underwriting on the next wave of project financings, lower advance rates on equipment facilities, ABS spread widening affecting insurer and fund buyers, and delayed data-center deliveries hurting equipment vendors. Each link is currently unbroken; the monitor’s job is to date the first break if one appears.

Circular financing. Documented loops this vintage: Nvidia’s \$105B guarantee enables SB Energy’s 4.25 GW buildout for an OpenAI affiliate, which will consume Nvidia GPUs, which collateralize the value of the guaranteed project; Microsoft’s \$13B funding of OpenAI alongside OpenAI’s Azure consumption commitments; Amazon’s \$8B Anthropic notes alongside the \$100B ten-year AWS commitment. These are disclosed commercial arrangements, mapped without treating circularity as wrongdoing. Quantification of the full loop (equity in, compute committed, financing raised, supplier revenue, collateral value) is not possible from current disclosure and is recorded as a gap.

8 Methodology, limitations and citations

Vintages. This is vintage 2026-09-27, append-only. Every stored value carries an as-of date and this vintage date. Revisions arrive as new vintages; history is never rewritten. The master table holds 70 entities, 90 obligations (57 primary, 33 memo), and 55 sources. One economic obligation gets one primary row; SPV instruments funding an already-recorded parent obligation are memo rows. Committed and drawn amounts are distinct facts; every aggregate states its basis. FX conversions use the as-of spot rate and record it.

What is unknown. The ranked gaps: (1) manager-level AI-infrastructure credit books outside BDC schedules; (2) OpenAI’s total debt, leases, purchase obligations, cash and covenants; (3) Anthropic’s balance sheet in full; (4) bank AI exposures beyond SMFG’s managerial-accounting figure; (5) incomplete tier ladders for Meta, Nvidia, Broadcom, Oracle, Nebius, Digital Realty and

Core Scientific; (6) covenant terms beyond the indentures extracted; (7) TTM interest expense, EBIT and operating cash flow gathered consistently, so interest coverage is not computed; (8) the lender-side refinancing calendar; (9) full circular-financing quantification; (10) FY2022 five-tier history beyond four borrowers' Tier A.

Paywall gaps. PitchBook, Preqin and S&P Capital IQ are inaccessible; public academic benchmarks (StepStone buyout dataset, Davis et al., Damodaran tables) stand in where comparable private-market multiples are needed. No paywalled figure is used.

Voice and checks. Dry precise research-lab voice. The mannerisms checklist was run against all prose; no em dashes appear anywhere, including charts and tables.

Key primary sources. Microsoft FY2026 10-K filed 29 Jul 2026 (Note 11); Alphabet 10-Q for quarter ended 30 Jun 2026 filed 23 Jul 2026 (Note 10); Amazon 10-Q filed 31 Jul 2026; Meta 10-Q filed 30 Jul 2026; Oracle FY2026 10-K filed 22 Jun 2026 and 10-Q filed 11 Sep 2026; Nvidia 10-Q filed 26 Aug 2026; Broadcom 10-Q filed 10 Sep 2026; CoreWeave 10-Q filed 12 Aug 2026 and 424B4 filed 31 Mar 2025; Nebius 6-K filed 12 Aug 2026; Applied Digital 10-K filed 29 Jul 2026; Core Scientific 10-Q filed 28 Jul 2026; TeraWulf 10-Q filed 5 Aug 2026; Equinix 10-Q filed 29 Jul 2026; Digital Realty 10-Q filed 31 Jul 2026; SMFG FY2025 investor presentation May 2026; bank Q2 2026 10-Qs and Pillar 3 disclosures as listed in `sources.csv`. Full citations with page, note or exhibit for every figure live in `data/vintages/2026-09-27/sources.csv`; the evidence narratives are in `evidence/cohortA_hyperscalers.md` through `cohortE_privatecredit.md`.